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Acne in Infancy and Acne Genetics

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Acne · Infancy · Genetics · Hereditary factors

Abstract
Acne is a disease that can be seen in the first year of age, early childhood, prepubertal age and puberty. Neonatal acne is due mainly to considerable sebum excretion rate, and infantile acne because of high androgens of adrenal origin in girls and of adrenal and testes in boys. These pathogenic mechanisms are characteristic in these ages. Important factors like early onset of comedones and high serum levels of dehydroepiandrosterone sulfate are predictors of severe or long-standing acne in prepubertal age. Hereditary factors play an important role in acne. Neonatal, nodulocystic acne and conglobate acne has proven genetic influences. Postadolescent acne is related with a first-degree relative with the condition in 50% of the cases. Chromosomal abnormalities, HLA phenotypes, polymorphism of human cytochrome P-450 1A1 and MUC1 gene are involved in the pathogenesis of acne. Several other genes are being studied.

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Neonatal Acne

Neonatal acne is present at birth or appears shortly after. It is more common than fully appreciated; if the diagnosis is based in a few comedones more than 20% of newborns are affected [1]. The most common lesions are comedones, papules and pustules. They are few in number and usually localized on the face, more often cheeks and forehead. Involvement of the chest, back or groins has been reported. Most cases are mild and transient. Lesions appear mainly at 2–4 weeks healing spontaneously, without scarring, in 4 weeks to 3–6 months. Neonatal acne has been suggested to be more frequent in male infants [2, 3].

The pathogenetic mechanisms of neonatal acne are still unclear. A positive family history of acne supports the importance of genetic factors. Familial hyperandrogenism including acne and hirsutism give the evidence that maternal androgens may play a role through transplacental stimulation of sebaceous glands [4]. There is a considerable sebum excretion rate during the neonatal period which decreases markedly to almost not detectable levels following the significant reduction of sebaceous gland volume up to the age of 6 months [5–7]. There is a direct correlation between high maternal and neonatal sebum excretion suggesting the importance of maternal environment on the infant sebaceous glands [8]. Neonatal adrenal glands produce a certain amount of β-hydroxysteroids that prepare the sebaceous glands to be more sensitive to hormones in the future life [1]. In males from 6 to 12 months there are increasing levels of luteinizing hormone (LH) and as a consequence of testosterone; these androgens plus those of testicular origin partially explain the male predominance of neonatal and infantile acne [3, 9].

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The differential diagnosis include milia, miliaria, sebaceous gland hyperplasia, bilateral naevus comedonicus, acneiform eruptions due to the use of topicals, oils and ointments, to maternal medications (lithium, hydantoin, steroids), or due to virilizing luteoma in pregnancy [1, 10, 11]. Deficiency of the 21-hydroxylase and adrenal cortical hyperplasia should also be considered [12]. Neonatal acne can also be confused with cephalic pustulosis due to malassezia species (mainly *Malassezia sympodialis*). Clinically the lesions are very similar to acne and are a consequence of an overgrowth of these lipophilic yeasts (on a neonate with high sebum production) that leads to an inflammatory reaction and poral or follicular occlusion. Its response to ketoconazole cream 2% is significant [13–15].

The treatment of neonatal acne begins with reassurance of parents. Topical treatments for comedones include retinoids such as tretinoin (cream 0.025–0.05%) or azelaic acid (cream 20%) daily or in alternating days. For inflammatory lesions, topical antibiotics (erythromycin 4% pads, pledgets, cream, gel) and benzoyl peroxide (wash, gel 2.5%) are useful [16].

Infantile Acne

Infantile acne (IA) usually starts later than neonatal acne, generally between 6 and 9 months (range 6–16 months) [16]. It also presents a male predominance. Lesions are localized on the face with the cheeks being the area most affected. A large survey on IA has been recently published showing that IA was mainly moderate in 62% of cases and mild and severe in 24% and 17% of cases, respectively. In addition to open and closed comedones, there were 59% of cases with inflammatory lesions and 17% with scars [17]. Occasional cases of conglobate acne can be seen; they occur primarily on the face and the clinical picture is exactly like the adult version.

The course is variable. Some cases disappear in 1 to 2 years but others are persistent and resolve at the age of 4–5 or persist until puberty.

Infantile acne, especially conglobate infantile acne, may be related with severe forms of the disease in adolescence. A family history of severe acne can be present [18].

The exact cause of IA is not clear. There is one case described with elevated levels of LH, follicle-stimulating hormone (FSH), and free testosterone due to an abnormality of the hypothalamus [19].

In severe cases of IA or persistent neonatal acne an infantile hyperandrogenemia should be excluded [16]. Physical examination looking for precocious puberty, bone age measurements and serological examinations FSH, LH, testosterone, dehydroepiandrosterone (DHEA) and dehydroepiandrosterone sulfate (DHEAS) are the initial approach. Any abnormality needs an endocrinologic evaluation.

Infantile acne must be differentiated from acneiform eruptions due to topical skin care products (greasy ointmemts, creams, pomades, oils) applied by the parents (pomade acne); due to steroids (topical, oral, inhaled) and from skin contact, ingestion or inhalation of aromatic hydrocarbons with chlorine groups (chloracne) [1, 20]. Perioral dermatitis can mimic an IA, papules and pustules are present mainly periorally (95%) and occasionally at the periocular area (44%). It can be associated to keratoconjunctivitis and vulvar lesions in female patients and usually occurs due to steroids. A family history is present in 20% of cases [21].

The treatment of IA in mild cases is with topical tretinoin, benzoyl peroxide or erythromycin. The main difficulties are the treatment of inflammatory lesions, deep papules and nodules that can persist for weeks or months. The oral antibiotics restricted to this age are erythromycin in doses of 125–250 mg twice daily and trimethoprim 100 mg twice daily in patients with shown resistance of *Propionibacterium acnes* to erythromycin [16, 17]. Deep nodules and cysts can be treated with an injection of low concentration of triamcinolone acetonide (2.5 mg/ml). If there is no response or nodular acne develops, which can lead to scarring, oral isotretinoin can be used. The doses proportionately are similar to adult (0.5 mg/kg/day for 4–5 months). Monitoring of complete and differential blood counts, liver function tests, cholesterol, triglyceride levels and a follow-up of skeletal involvement should be performed [22, 23].

Parents have to be informed that the treatment is a long-term one with possibilities of reappearance of acne at puberty.

Mid-Childhood Acne

This type of acne occurs between 1 and 7 years of age. Acne is very rare in this group and when it occurs should be evaluated for hyperandrogenemia.

Differential diagnosis includes Cushing's syndrome, congenital adrenal hyperplasia, gonadal or adrenal tumors and a true precocious puberty. Evaluation should be

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done with a bone age measurement, growth chart and laboratory tests that include serum total and free testosterone, DHEA, DHEAS, LH, FSH, prolactin and 17α-hydroxyprogesterone. Occasional reports of acne at this age because of *D*-actinomycin are available in the literature.

Mid-childhood acne can be confused sometimes with keratosis pilaris of the cheeks and with keratin cysts (milia) particularly when they get inflamed. Both lesions are common in atopics [3, 16].

The therapy is identical to that of infantile acne.

Prepubertal Acne

Increasing number of early onset acne before obvious signs of puberty is a recognized phenomenon associated more with pubertal development than with age. There is apparently a genetic predisposition.

Pubertal development has two components, normal adrenarche related to maturation of adrenal glands and true puberty because of maturation of testis and ovary mediated by the hypothalamic-pituitary axis.

Adrenarche presents with high levels of DHEA and DHEAS that start rising at 6–7 years in girls and 7–8 years in boys and follow increasing during mid puberty. Excessive androgen production may result due to adrenal hyperandrogenism (exaggerated adrenarche, exuberant production of adrenal androgens relative to cortisol), congenital adrenal hyperplasia, Cushing's disease, 21-hyroxylase deficiency, and more rarely androgen producing tumors. Ovarian contribution to androgens can be through tumors (malignant and benign), but most commonly due to polycystic ovarian disease associated very often with obesity, persistent or resistant acne and insulin resistance [3, 12].

Acne could be the first sign of pubertal maturation and associated with increase in sebum and urinary excretion of androgenic steroids. A high frequency of acne was found in a longitudinal study of adolescent boys, where the prevalence and severity of acne correlated well with advanced pubertal maturation [24]. A similar study of the same authors in early adolescent girls concluded that acne can be the first sign of pubertal maturation; significant elevations of DHEAS correlated well in girls with comedonal and inflammatory acne. The most common locations of acne in this group were the midforehead, nose and chin [25].

In a longitudinal study of acne and hormonal analysis, Stewart et al. [26] concluded that girls with severe acne present a statistically significantly earlier menarche (12.2 years) compared to those with moderate and mild disease (12.4 and 12.7 years). They also concluded that the number of comedones were predictive for the severity of late inflammatory acne. Mid-pubertal girls with severe comedonal acne showed more comedones even three years before menarche. This group also showed higher levels of DHEAS early in life. A correlation between DHEAS, sebum production and free testosterone was found in severe comedonal acne [26].

Lucky et al. [27] in a 5-year longitudinal cohort study of 871 girls stated clearly the predictor factors of an acne vulgaris study. They evaluated acne versus hormone levels at various ages before and after menarche. They were able to conclude that there were no ethnic differences in acne or hormone levels in the groups studied that included black and white girls. A progressive increase in number of acne lesions with age and maturation was found. The most common acne was comedonal; girls with severe acne at the end of the study had more comedones and inflammatory lesions by the age of 10 years and 2.5 years before menarche. The onset of menarche was also earlier in cases with severe acne and associated to higher levels of serum DHEAS and total and free testosterone compared to girls with mild-to-moderate disease. Early development of comedonal acne, DHEAS, free and total testosterone were good predictors for severe comedonal acne or a long-term disease [27].

The differential diagnosis is essentially similar to midchilhood acne. Adverse effects of certain drugs (corticosteroids, anticonvulsants, etc.) and sporadic cases of prepubertal hydradenitis suppurative must be considered [28].

The therapy of acne at this age is similar to that reported before. Topical retinoids, benzoyl peroxide, antibiotics are appropiate in mild-to-moderate comedonal and inflammatory acne. In more severe cases, especially in risk of scarring, the use of oral antibiotics and oral isotretinoin may be necessary. Resistant, persistent and cases of acne appearing at unusual ages need hormonal evaluation and proper treatment. Adrenal problems may need low doses of oral corticosteroids; polycystic ovarian disease can be treated with oral contraceptives that include antiandrogens such as cyproterone acetate or the new low androgenic progestins. Spironolactone can also be considered [29].

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Acne Genetics

The genetic influence on pathogenesis of acne is well documented in twins [30] and genealogic studies. In some types of acne, such as acne conglobata, hereditary factors are more apparent, and a correlation has been suggested between neonatal acne and familial hyperandrogenism [4]. Nodulocystic IA is often observed in relatives of patients with extensive steatocystoma, adolescent and postadolescent acne [31]. Fifty percent of postadolescent acne patients have at least one first-degree relative with the condition [32].

Sebum excretion also correlates with acne susceptibility, and sebum excretion rates are similar in identical twins [33]. Several chromosomal abnormalities, including 46XYY genotype [34], 46XY+ (4p+; 14q–) [35], and partial trisomy 13 [36] have been reported to be associated with nodulocystic acne.

The relationship of acne and various genes has been investigated. An HLA antigen study was negative for acne conglobata [37], but HLA phenotypes were identical in siblings affected with familial acne fulminans [38]. Polymorphism in the human cytochrome P-450 1A1 (CYP1A1) seem to be associated with acne [39], and CYP1A1 is known to be involved in the metabolism of a wide range of compounds such as vitamin A. In acne patients, a higher frequency of CYP1A1 mutation was observed on regulatory sites, and this may impair the biological efficacy of natural retinoids due to their rapid metabolism to inactive compounds. This mutation may thus be involved in the pathogenesis of acne in some patients. CYP1A1 inducibility is determined by polymorphism in the genes of the aromatic hydrocarbon (Ah) receptor. This Ah receptor mediates the toxic effects of environmental pollutants such as dioxin and polyhalogenated biphenyls. Clinical correlations between the high inducibility of CYP1A1 and some carcinomas are observed; however, no correlation was found between polymorphism of the human Ah receptor and 2,3,7,8-tetrachlorodibenzo-*p*-dioxin-induced chloracne in chemical workers accidentally exposed to this chemical [40].

An inadequate activity of steroid 21-hydroxylase, as well as CYP21 gene mutations, is the genetic basis for congenital or late-onset adrenal hyperplasia which may present with acne. Acne patients exhibit a high frequency of a CYP21 gene mutation, but a poor correlation exists between mutations and either elevated steroids or acne [41]. It has been suggested that factors other than mild impairment of CYP21 can contribute to the clinical phenotype that includes acne.

Androgen receptor polymorphisms of CAG trinucleotide repeat length has clinical implications for human disease. This polymorphism exhibits a correlation with some androgenic skin diseases but not with acne [42].

MUC1 is a glycoprotein secreted from various epithelial glands including sebaceous glands. Studies of the respiratory and digestive systems suggest that MUC1 is involved in the defense system against bacteria by inhibiting their adhesion to epithelium. The MUC1 gene and the molecule produced exhibit extensive polymorphism attributable to a variable number of tandem repeats. A higher frequency of longer repeat length of tandem repeats has been observed in severe acne patients [43].

The melanocortin 5 receptor is known to regulate sebaceous gland function in mouse. Genetic diversity is observed in human melanocortin 5 receptor coding region. Association between variation at the locus and acne is not found [44].

In conclusion, the pathogenesis of acne is multifactorial and a greater number of genes than those cited above are probably related to the condition. Genes affecting keratinization and desquamation are suspected to be involved in the pathogenesis of acne and their correlation to acne is yet to be evaluated. Advances in immunogenetic research may shed new light on the understanding of the inflammatory reaction in acne. Genes expressed in the sebaceous glands which exhibit polymorphism are of special interest, regardless of their known function. Any gene polymorphisms found to be related to acne may provide additional insights into the pathogenesis of this condition. Further research is needed to investigate the combined effects of these and other genes.

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References

1 Jansen Th, Burgdorf W, Plewig G: Pathogenesis and treatment of acne in childhood. Pediatr Dermatol 1997;14:7–21.

2 Katsambas AD, Katoulis AC, Stavropoulos P: Acne neonatorum: A study of 22 cases. Int J Dermatol 1999;38:128–130.

3 Lucky AW: A review of infantile acne and pediatric acne. Dermatology 1998;196:95–97.

4 Bekaert C, Song M, Delvigne A: Acne neonatorum and familial hyperandrogenism. Dermatology 1998;196:453–454.

5 Agache P, Blanc D, Laurent R: Sebum levels during the first year of life. Br J Dermatol 1980; 103:643–649.

6 Zouboulis CC, Fimmel S, Ortmann J, Turnbull J, Boschnakow A, Pochi P: Sebaceous Glands; in Hoath SB, Maibach H (eds): Neonatal Skin-Structure and Function, ed 2. New York, Marcel Dekker, in press.

7 Zouboulis CC, Boschnakow A: Chronological ageing and photoageing of the human sebaceous gland. Clin Exp Dermatol 2001;26:600–607.

8 Henderson CA, Taylor J, Cunliffe WJ: Sebum excretion rates in mothers and neonates. Br J Dermatol 2000;142:110–111.

9 Rosenfield RL: Hyperandrogenism in peripubertal girls. Pediatr Clin N Am 1990;37:1333–1358.

10 Vasiloudes PE, Morelli JG, Weston WL: Inflammatory nevus comedonicus in children. J Am Acad Dermatol 1998;38:834–836.

11 Janniger CK: Neonatal and infantile acne vulgaris. Cutis 1993;52:16.

12 De Raeva L, De Schepper J, Smitz J: Prepubertal acne: Cutaneous marker of androgen excess? J Am Acad Dermatol 1995;32:181–184.

13 Van Praag MCG, Van Rooij RWG, Folkers E, Spritzer R, Henke HE, Oranje AP: Diagnosis and treatment of pustular disorders in the neonate. Pediatr Dermatol 1997;14:131–134.

14 Rappelanoro R, Morturerx P, Couprie B, Maleville J, Taieb A: Neonatal *Malassezia furfur* pustulosis. Arch Dermatol 1996;132:190–193.

15 Niamba P, Weill FX, Sarlangue J, Labreze C, Couprie B, Taieb A: Is common neonatal cephalic pustulosis (neonatal acne) triggered by *Malassezia sympodialis*? Arch Dermatol 1998; 134:995–998.

16 Lucky AW: Acne therapy and childhood. Dermatol Ther 1998;6:74–81.

17 Cunliffe WJ, Baron SE, Coulson IH: A clinical and therapeutic study of 29 patients with infantile acne. Br J Dermatol 2001;145:463–466.

18 Chew EW, Bingham A, Burrows D: Incidence of acne vulgaris in patients with infantile acne. Clin Exp Dermatol 1990;15:376–377.

19 Duke EMC: Infantile acne associated with transient increases in plasma concentrations of luteinizing hormone, follicle-stimulating hormone and testosterone. Akt Dermatol 1987;13: 306–307.

20 Caputo R, Monti M, Ermacora E, Carminati G, Gelmetti C, Gianotti R, Gianni E, Puccinelli V: Cutaneous manifestations of tetrachlorodibenzo-*P*-dioxin in children and adolescents: Follow-up 10 years after the Seveso, Italy, accident. J Am Acad Dermatol 1988;19:812–819.

21 Manders S, Lucky AW: Perioral dermatitis in childhood. J Am Acad Dermatol 1992;27:688–692.

22 Arbegast KD, Braddock SW, Lamberty LF, Sawka AR: Treatment of infantile cystic acne with oral isotretinoin: A case report. Pediatr Dermatol 1991;8:166–168.

23 Horne HJ, Carmichael AJ: Juvenile nodulocystic acne responding to systemic isotretinoin. Br J Dermatol 1997;136:796–797.

24 Lucky AW, Biro FM, Huster GA, Morrison JA, Elder N: Acne vulgaris in early adolescent boys. Arch Dermatol 1991;127:210–216.

25 Lucky AW, Biro FM, Huster GA, Leach AD, Morrison JA, Ratterman J: Acne vulgaris in premenarchal girls: An early sign of puberty associated with rising levels of dehydroepiandrosterone. Arch Dermatol 1994;130:308–314.

26 Stewart ME, Downing DT, Cook JS, Hansen JR, Strauss JS: Sebaceous gland activity and serum dehydroepiandrosterone sulfate levels in boys and girls. Arch Dermatol 1992;128:1345–1348.

27 Lucky AW, Biro FM, Simbartl LA, Morrison JA, Sorg NW: Predictors of severity of acne vulgaris in young adolescent girls: Results of a five year longitudinal study. J Pediatr 1997; 130:30–39.

28 Nengesha YM, Holcombe TC, Hansen RC: Prepubertal hidradenitis suppurativa: Two case reports and review of the literature. Pediatr Dermatol 1999;16:292–296.

29 Lucky AW, Henderson TA, Olson WH, Robisch DM, Lebwohl M, Swinyer LJ: Effectiveness of norgestimate and ethinyl estradiol in treating moderate acne vulgaris. J Am Acad Dermatol 1997;37:746–754.

30 Friedman GD: Twin studies of disease heritability based on medical records: application to acne vulgaris. Acta Genet Med Gemellol 1984; 33:487–495.

31 Burket JM, Storrs FJ: Nodulocystic infantile acne occurring in a kindred of steatocystoma. Arch Dermatol 1987;123:432–433.

32 Goulden V, Clark SM, Cunliffe WJ: Post-adolescent acne: a review of clinical features. Br J Dermatol 1997;136:66–70.

33 Walton S, Wyatt Eh, Cunliffe WJ: Genetic control of sebum excretion and acne: A twin study. Br J Dermatol 1998;118:393–396.

34 Voorhees JJ, Wilkins JW; Hayes E, Harrell R: Nodulocystic acne as a phenotypic feature of the XYY genotype. Arch Dermatol 1972;105: 913–919.

35 Marr TJ, Traisman HS: Nodulocystic acne, Chromosomal abnormality, and diabetes mellitus. Cutis 1981;27:87–88.

36 Funderburk SJ, Landau JW: Acne in retarded boy with autosomal chromosomal abnormality. Arch Dermatol 1976;112:859–861.

37 Schackert K, Scholz S, Steinbauer-Rosenthal I, Albert ED, Wank R, Plewig G: HLA antigen in acne conglobata: A negative study. Arch Dermatol 1974;110:468.

38 Wong SS, Pritchard MH, Holt PJA: Familial acne fulminans. Clin Exp Dermatol 1992;17: 351–353.

39 Paraskevaidis A, Drakoulis N, Roots I, Orfanos CE, Zouboulis CC: Polymorphisms in the human cytochrome P-450 1A1 gene (CYP1A1) as a factor for developing acne. Dermatology 1998;196:171–175.

40 Wanner R, Zober A, Abraham K, Kleffe J, Henz BM, Wittig B: Polymorphism at codon 554 of human Ah receptor: Different allelic frequencies in Caucasians and Japanese and no correlation with severity of TCDD-induced chloracne in chemical workers. Pharmacogenetics 1999;9:777–780.

41 Ostlere LS, Rumsby G, Holownia P, Jacobs HS, Rustin MHA, Honour JW: Carrier status for steroid 21-hydroxylase deficiency is only one factor in the variable phenotype of acne. Clin Endocrinol 1998;48:209–215.

42 Sawaya ME, Shalita AR: Androgen receptor polymorphisms (CAG repeat length) in androgenetic alopecia, hirsutism, and acne. J Cutan Med Surg 1998;3:9–15.

43 Ando I, Kukita A, Soma G, Hino H: A large number of tandem repeats in the polymorphic epithelial mucin gene is associated with severe acne. J Dermatol 1998;25:150–152.

44 Hatta N, Dixon C, Ray AJ, Phillips SR, Dale M, Todd C, Meggit S, Birch-Machin MA, Rees JL: Expression, candidate gene, and population studies of the melanocortin 5 receptor. J Invest Dermatol 2001;116:564–570.

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